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The Sensitivity of Eight Years of the Zwicky Transient Facility to Kilonovae: Injection–Recovery Constraints on the Kilonova Rate

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ABSTRACT

We quantify the sensitivity of the Zwicky Transient Facility (ZTF) to optical kilonovae over its first eight years of operations (2018 March – 2026 March; $T = 8.006\,\mathrm{yr}$) by injecting 10^8 synthetic kilonovae per model into the ZTF exposure history and applying fast-transient selection criteria similar to those of real-time ZTF kilonova searches. For kilonovae with AT2017gfo-like ejecta (two-component POSSIS models evaluated with the `fiesta` surrogate) viewed at $\iota = 25.8°$, the effective volume–time is $\langle VT \rangle = 0.0157\,\mathrm{Gpc^3\,yr}$; if no candidate in the ZTF data passes the same criteria, the 95% confidence upper limit on the rate is $\mathcal{R}_{95} = 191\,\mathrm{Gpc^{-3}\,yr^{-1}}$. For the same ejecta with isotropically distributed viewing angles, $\langle VT \rangle = 0.0048\,\mathrm{Gpc^3\,yr}$ and $\mathcal{R}_{95} = 624\,\mathrm{Gpc^{-3}\,yr^{-1}}$. The isotropic sensitivity is a factor 3.3 lower because recovery efficiency falls by more than two orders of magnitude between polar and equatorial viewing angles; injections with $\iota > 45°$ make up 71% of the isotropic population but only 19% of the recovered sample. Restricting our injections to the first 23 months of ZTF gives $\mathcal{R}_{95} = 1036\,\mathrm{Gpc^{-3}\,yr^{-1}}$ (polar view), and the full survey accumulates 5.4× more volume–time than that window, more than the 4.2× expected from duration alone. Monte Carlo uncertainties are below 0.5%; the limits are dominated by model and selection systematics, which we discuss. [TBD: Population-averaged limit (Suite 03 re-run), real-data candidate search result, and criteria-systematics band (Suite 02 re-run).]

Keywords: Gravitational wave sources (677) — Neutron stars (1108) — Transient sources (1851) — Sky surveys (1464)

1. INTRODUCTION

The radioactive decay of r-process nuclei synthesized in the neutron-rich ejecta of binary neutron star (BNS) and neutron star–black hole mergers powers a thermal optical/near-infrared transient, the kilonova (Li & Paczyński 1998; Metzger et al. 2010; Roberts et al. 2011; Barnes & Kasen 2013; Kasen et al. 2013; Tanaka & Hotokezaka 2013; Metzger 2020). AT2017gfo, the counterpart of GW170817 at $\approx 40\,\mathrm{Mpc}$ (Abbott et al. 2017a,b; Coulter et al. 2017; Soares-Santos et al. 2017; Arcavi et al. 2017), reached peak absolute magnitudes $M_g \approx M_r \approx -16.0$ and faded by $\approx 1\,\mathrm{mag\,d^{-1}}$ in g and r (Drout et al. 2017; Cowperthwaite et al. 2017).

Wide-field optical surveys can find kilonovae independently of gravitational-wave triggers. Searches with no detections have constrained the rate of AT2017gfo-like events: $\mathcal{R} < 800\,\mathrm{Gpc^{-3}\,yr^{-1}}$ from galaxy-targeted Palomar Transient Factory data (Kasliwal et al. 2017), $< 3.0 \times 10^4\,\mathrm{Gpc^{-3}\,yr^{-1}}$ from ATLAS (Smartt et al. 2017), and $< 2.4 \times 10^4\,\mathrm{Gpc^{-3}\,yr^{-1}}$ (90%) for the brightest model in the Dark Energy Survey (Doctor et al. 2017). With the first 23 months of ZTF (Bellm et al. 2019; Graham et al. 2019; Dekany et al. 2020; Masci et al. 2019), Andreoni et al. (2020) found $\mathcal{R} < 1775\,\mathrm{Gpc^{-3}\,yr^{-1}}$ (95%) for GW170817-like kilonovae and $\mathcal{R} < 4029\,\mathrm{Gpc^{-3}\,yr^{-1}}$ for POSSIS models with uniformly distributed viewing angles; the real-time ZTFReST search added $\mathcal{R} < 1904\,\mathrm{Gpc^{-3}\,yr^{-1}}$ and $< 900\,\mathrm{Gpc^{-3}\,yr^{-1}}$ combined (Andreoni et al. 2021). For comparison, the BNS merger rate from gravitational waves is 7.6–250 $\mathrm{Gpc^{-3}\,yr^{-1}}$ (90%; GWTC-4.0, Abac et al. 2026), revised from 10–1700 $\mathrm{Gpc^{-3}\,yr^{-1}}$ in GWTC-3 (Abbott et al. 2023).

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Here we compute the sensitivity of the full eight-year ZTF data set to kilonovae with an injection–recovery simulation and give the rate limits that follow if no candidate survives the selection. We describe the survey and criteria in Section 2, the simulation and statistics in Section 3, the limits and their time dependence in Section 4, the viewing-angle dependence in Section 5, Monte Carlo convergence in Section 6, and systematics in Section 8. [TBD: Section on the archival candidate search (Section 7).]

2. ZTF DATA AND SELECTION CRITERIA

ZTF observes with a 47 deg^2 camera on the Palomar 48-inch Schmidt telescope in g, r and i (Bellm et al. 2019; Dekany et al. 2020). We use the exposure history from 2018 March to 2026 March (97 calendar months; $T = 8.006$ yr from the first to the last exposure). For each exposure the simulation uses the pointing, filter, epoch and 5σ limiting magnitude of the difference image (median over the CCD quadrants; Masci et al. 2019).

An injected source counts as recovered if (i) at least two exposures reach $S/N \geq 3$, at least one of which reaches $S/N \geq 5$; (ii) the first and last such detections are separated by at least 3 h and at most 14 d; (iii) the source fades by at least 0.3 mag d^{-1} in at least one band, measured as a weighted least-squares slope of the post-peak detections in that band (same-band points within 2 h are averaged); and (iv) it lies at Galactic latitude $|b| \geq 15°$. These criteria resemble those of ZTFReST (≥ 2 detections at $> 3\sigma$ with one at $> 5\sigma$, separated by > 3 h, fading > 0.3 mag d^{-1}, $|b| > 10°$; Andreoni et al. 2021); Andreoni et al. (2020) instead required band-dependent fade rates of 0.57/0.39/0.27 mag d^{-1} ($g/r/i$) and applied no Galactic-latitude cut. The sensitivity of $\langle VT \rangle$ to each criterion is the subject of a companion paper [TBD: companion citation].

3. SIMULATION AND STATISTICS

3.1. *Injections*

We use the `survey-sim` injection–recovery code [TBD: citation/software DOI]. Injected kilonovae are distributed isotropically over the full sky, uniformly in comoving volume out to $z_{\rm max} = 0.15$, and uniformly in explosion time over the survey. Each injection is evaluated against the ZTF exposures that cover its position from 30 d before to $25(1+z)$ d after explosion, using the noiseless model magnitude and the exposure's limiting magnitude ($S/N = 5 \times 10^{0.4(m_5 - m)}$). We assume a flat ΛCDM cosmology with $h = 0.674$ and $\Omega_m = 0.315$ (Planck Collaboration et al. 2020). The 99th percentile of the recovered redshifts is $z = 0.059$ ($d_L = 274$ Mpc) for the polar-view POSSIS model. For every model except the one-zone "blue" template, the most distant recovery is at $z \leq 0.105$, so $z_{\rm max}$ does not truncate $\langle VT \rangle$ (Figure 3). Recoveries of the "blue" template extend to $z_{\rm max}$, so its $\langle VT \rangle$ is a lower bound.

3.2. *Effective volume–time and rate limits*

For a model with recovery efficiency $\epsilon(z)$,

$$\langle VT \rangle = 4\pi\, T \int_0^{z_{\rm max}} \epsilon(z)\, \frac{dV_c}{dz\, d\Omega}\, \frac{1}{1+z}\, dz, \qquad (1)$$

where the factor $1/(1+z)$ converts source-frame to observer-frame rates. Because injections are full-sky, the survey footprint, observing gaps and the latitude cut are all contained in ϵ. Each model is simulated in 100 independent realisations of 10^6 injections with distinct random seeds. We combine them by injection-weighted averaging of $\langle VT \rangle$, which is the minimum-variance estimate, rather than by averaging per-realisation limits. With $N_{\rm obs}$ real candidates and no background, the classical upper limit at confidence level CL is $\mathcal{R}_{\rm CL} = \mu_{\rm CL}(N_{\rm obs})/\langle VT \rangle$, with $\mu_{95}(0) = -\ln 0.05 = 2.996$; the Feldman–Cousins interval would give 3.09 (Feldman & Cousins 1998). [TBD: $N_{\rm obs}$ from the archival search; the limits below assume $N_{\rm obs} = 0$.]

3.3. *Kilonova models*

1. **POSSIS two-component models.** We use the `fiesta` neural-network surrogate (Koehn et al. 2025) of POSSIS radiative-transfer light curves (Bulla 2019, 2023), parameterised by the mass, velocity and electron fraction of the dynamical and wind ejecta and the viewing angle ι from the pole. Two parameter sets are used, which we call the baseline set ($\log_{10}(M_{\rm dyn}/M_\odot) = -1.7$) and the GW170817 set ($-1.8$), both with $v_{\rm dyn} = 0.2c$, $Y_{e,\rm dyn} = 0.15$, $\log_{10}(M_{\rm wind}/M_\odot) = -1.1$, $v_{\rm wind} = 0.1c$ and $Y_{e,\rm wind} = 0.35$. Each is simulated at fixed $\iota = 0.45$ rad (25.8°) and with $\cos\iota$ uniform in $[0, 1]$. The viewing angle of GW170817 is 14°–28° from radio VLBI (Mooley et al. 2018) and 19°–25° including optical astrometry (Mooley et al. 2022). [TBD: Replace the ad hoc parameter set by a published AT2017gfo posterior (e.g. Anand et al. 2023) and marginalise over it.]

2. **One-zone (Metzger-type) models.** [TBD: Likely removed: the literature-preset models are not calibrated and the AT2017gfo template is kept only if it matches AT2017gfo photometry (see docs/paper_revision_plan.md).] These are included for comparison with earlier work (literature-based "blue" and "mean" parameters:

$M_{\rm ej} = 0.02\,M_\odot$, $v = 0.25c$, $\kappa = 0.5$ and $M_{\rm ej} = 0.04\,M_\odot$, $v = 0.15c$, $\kappa = 1$; and a template fitted to AT2017gfo photometry, $M_{\rm ej} = 1.26\times10^{-3}\,M_\odot$, $v = 0.5c$, $\kappa = 398$). In this implementation κ is not the grey opacity of the ejecta, and the parameters are not directly comparable to the component fits of Villar et al. (2017). The recovered "blue" model reaches a median observed $M \approx -17.6$, about 1.6 mag brighter than AT2017gfo, so we do not treat it as AT2017gfo-like. [TBD: Re-tune or drop the one-zone models.]

3. **Population.** [TBD: Suite 03 re-run: per-transient parameter draws (`Bu2026KilonovaPopulation`) and isotropic ι; prior choice to be justified against numerical-relativity ejecta predictions, e.g. Radice et al. 2018.]

4. RESULTS

Table 1 lists the pooled $\langle VT\rangle$ and the 90/95/99% limits for all fixed-parameter models, and Figure 1 compares them with previous searches and the gravitational-wave BNS rate. For AT2017gfo-like ejecta viewed at 25.8° the two parameter sets give $\mathcal{R}_{95} = 192$ (baseline set) and 191 $\mathrm{Gpc^{-3}\,yr^{-1}}$ (GW170817 set). These differ by 2.4σ of the Monte Carlo error, consistent with the 0.1 dex change in $M_{\rm dyn}$. For isotropic viewing angles, $\mathcal{R}_{95} = 664$ and 624 $\mathrm{Gpc^{-3}\,yr^{-1}}$.

4.1. *Accumulation of sensitivity over the survey*

Because explosion times are uniform over the survey, the volume–time from explosions in any time window equals $\langle VT\rangle$ times the fraction of recovered injections in that window. Figure 2 shows that ZTF's monthly sensitivity varies by more than a factor of two around its mean and nearly vanishes in 2021 December – 2022 January. Restricting the injections to the first 23 calendar months (2018 March – 2020 January) gives $\mathcal{R}_{95} = 1036\,\mathrm{Gpc^{-3}\,yr^{-1}}$ for the polar-view model (GW170817 set) and 3416 $\mathrm{Gpc^{-3}\,yr^{-1}}$ for isotropic viewing. The full survey accumulates 5.4–5.8× more $\langle VT\rangle$ than this window, compared with a duration ratio of 4.2. Our 23-month limits are ≈ 1.7× (polar) and 1.2× (isotropic) tighter than those of Andreoni et al. (2020), which used different criteria, extinction treatment and a real-data search. A limit scales as $1/\langle VT\rangle$, so carrying a published limit to the full survey requires the growth of $\langle VT\rangle$ for that search; a scaling with survey duration alone would understate it, because ZTF accumulated $\langle VT\rangle$ faster after the first 23 months than during them. Using instead the growth we measure, and assuming that the ratio between the published efficiency and ours is the same in every epoch, the Andreoni et al. (2020) limits would correspond to ≈ 320 (polar) and ≈ 730 $\mathrm{Gpc^{-3}\,yr^{-1}}$ (isotropic) for eight years of data (Figure 2, dashed). [TBD: Quantify how much of this difference comes from criteria (Suite 02) and from pipeline losses (Section 8).]

5. DEPENDENCE ON VIEWING ANGLE

The isotropic ensembles record the viewing angle of every recovered injection. Because the injected $\cos\iota$ is uniform, the recovery efficiency in each $\cos\iota$ bin is exact, and so is the $\langle VT\rangle$ a population fixed at that angle would have (Figure 4). For the baseline set, $\langle VT\rangle(\iota)$ falls from 0.022 $\mathrm{Gpc^3\,yr}$ ($\mathcal{R}_{95} = 135\,\mathrm{Gpc^{-3}\,yr^{-1}}$) for $\iota < 18°$ to 7×10^{-5} $\mathrm{Gpc^3\,yr}$ ($\mathcal{R}_{95} \approx 4\times10^4\,\mathrm{Gpc^{-3}\,yr^{-1}}$) for $\iota > 87°$. Injections with $\iota > 45°$ make up 71% of an isotropic population but only 16% of the recovered sample (19% for the GW170817 set). As an internal check, interpolating $\langle VT\rangle(\iota)$ to $\iota = 0.45$ rad reproduces the independent fixed-angle ensemble to 0.03% (baseline set) and 1.1% (GW170817 set). Recovered polar sources also reach larger distances than equatorial ones (Figure 4, right).

6. MONTE CARLO CONVERGENCE

We estimate the Monte Carlo precision by subsampling the actual realisations: the number of recoveries in a random subset of N injections is hypergeometric given a realisation's totals, and larger N are built from sums of distinct realisations. The scatter of $\langle VT\rangle$ follows the binomial expectation $\sqrt{(1-\epsilon)/(N\epsilon)}$ at all N (Figure 5). It is 2.3% (polar POSSIS) and 4.5% (isotropic) for a single 10^6-injection realisation, and 0.2% and 0.45% for the pooled 10^8. For the isotropic model, a 10^4-injection run has a 0.4% chance of no recoveries. The run-to-run scatter of the 100 realisations agrees with binomial noise to within 7%.

7. ARCHIVAL CANDIDATE SEARCH

[TBD: Apply the selection of Section 2 to ZTF alert and forced photometry for 2018–2026, including real-bogus, stellar and Solar-System vetoes as in Andreoni et al. (2020, 2021); report $N_{\rm obs}$, the vetting of survivors, and the efficiency of the vetting steps applied to injected light curves. Known imposter classes are catalogued by Barna et al. (2025).]

8. SYSTEMATIC UNCERTAINTIES

- **Real-data pipeline losses.** The simulation does not include real-bogus classification (Duev et al. 2019; Mahabal et al. 2019), stellar or minor-planet vetoes, or bad subtractions. These remove genuine

Table 1. Pooled effective volume–time and Poisson upper limits (assuming $N_{\rm obs} = 0$) for the fixed-parameter models, each from 10^8 injections. Rates are in $\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$; the $\langle VT\rangle$ uncertainty is the Monte Carlo standard error.

Model	Parameters	$N_{\rm rec}$	$\langle VT\rangle$ [10^{-3} Gpc^3 yr]	$\mathcal{R}_{90}$	$\mathcal{R}_{95}$	$\mathcal{R}_{99}$
Bu2026, $\iota = 26°$	$\log M_{\rm dyn} = -1.7$, $\log M_{\rm wind} = -1.1$, $\iota = 25.8°$	184316	15.60 ± 0.04	148	192	295
Bu2026, isotropic ι	$\log M_{\rm dyn} = -1.7$, $\log M_{\rm wind} = -1.1$, isotropic	53948	4.51 ± 0.02	510	664	1020
Metzger, $\kappa = 398$	$M_{\rm ej} = 0.00126\,M_\odot$, $v = 0.5c$, $\kappa = 398$	47831	3.96 ± 0.02	581	756	1162
Bu2026, $\iota = 26°$	$\log M_{\rm dyn} = -1.8$, $\log M_{\rm wind} = -1.1$, $\iota = 25.8°$	186028	15.72 ± 0.03	146	191	293
Bu2026, isotropic ι	$\log M_{\rm dyn} = -1.8$, $\log M_{\rm wind} = -1.1$, isotropic	57435	4.80 ± 0.02	479	624	959
Metzger blue, $\kappa = 0.5$	$M_{\rm ej} = 0.02\,M_\odot$, $v = 0.25c$, $\kappa = 0.5$	1316108	110.47 ± 0.09	21	27	42
Metzger mean, $\kappa = 1$	$M_{\rm ej} = 0.04\,M_\odot$, $v = 0.15c$, $\kappa = 1$	468311	39.77 ± 0.05	58	75	116

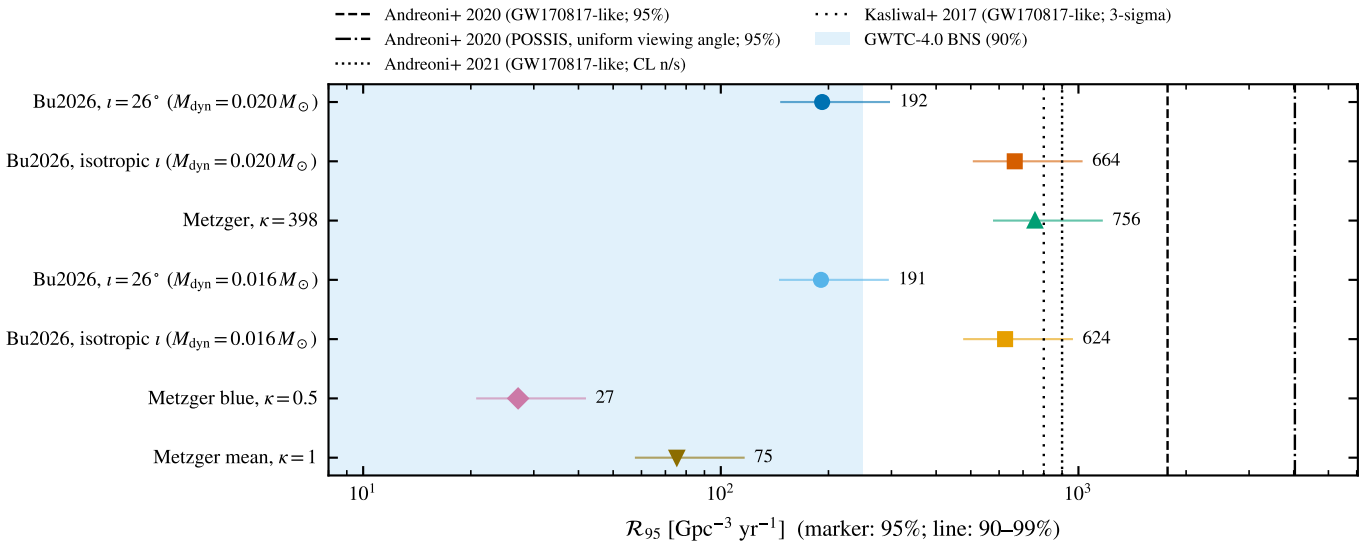


Figure 1. 95% upper limits for each model (markers; error bars are the Monte Carlo uncertainty, thin lines span the 90%–99% limits). Vertical lines show published optical limits: Andreoni et al. (2020) for GW170817-like and POSSIS isotropic models, the ZTFReST combined limit (Andreoni et al. 2021) and the PTF limit (Kasliwal et al. 2017). The shaded band is the GWTC-4.0 BNS merger rate (90%; Abac et al. 2026).

transients, so ϵ is an upper bound on the efficiency of a real search and the limits are correspondingly optimistic. [TBD: Measure by injecting into the real vetting chain, or adopt published efficiencies with uncertainties.]

- **Extinction.** No Milky Way or host extinction is applied to the POSSIS models; the one-zone models include a constant $A_V = 0.02$ and a host term. [TBD: Apply position-dependent dust maps to all models.]

- **Photometric noise.** Detection uses noiseless model magnitudes, so the recovery probability near threshold is idealised.

- **Surrogate time axis.** In the present version of the `fiesta` interface the light curves are stretched by $(1+z)^2$ instead of $(1+z)$; correcting this lowers $\langle VT\rangle$ of the POSSIS models by $\approx 4\%$. [TBD: Rerun after the fix.]

- **Fast-transient test.** The fade requirement removes $\approx$ 24% of otherwise recovered polar-view injections. With a rise threshold of zero, any detected rise also satisfies the test in the current implementation, which changes $\langle VT\rangle$ by $\approx 1\%$.

- **Numerical integration.** Evaluating Eq. 1 on 20 redshift bins underestimates $\langle VT\rangle$ by 2%–5% compared with a direct Monte Carlo sum. This makes the limits conservative.

- **Model choice.** The fixed-parameter limits differ by $\approx$ 3.3–3.5$\times$ between polar and isotropic viewing and by an order of magnitude across the one-zone models [TBD: drop this clause if the one-zone models are removed]. This is far larger than the Monte Carlo error. [TBD: Population and posterior-marginalised limits.]

9. DISCUSSION AND CONCLUSIONS

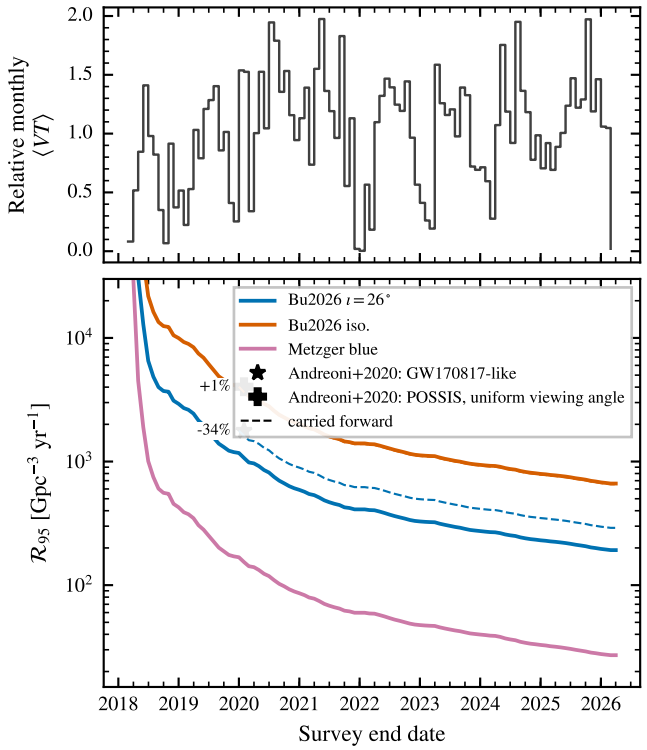


Figure 2. Top: relative monthly sensitivity (volume–time contributed by each calendar month of explosions, normalised to the mean) for the polar-view POSSIS model. Bottom: cumulative 95% limit as a function of survey end date for three models, with the Andreoni et al. (2020) limits plotted at the end of their 23-month window, labelled with the change of our limit relative to theirs over the same window, and carried forward with the measured growth of $\langle VT \rangle$ (dashed).

If no kilonova candidate in the eight-year ZTF data passes our criteria, the 95% limits are $\mathcal{R}_{95} = 191\,\mathrm{Gpc^{-3}\,yr^{-1}}$ for AT2017gfo-like ejecta viewed at 25.8° and $624\,\mathrm{Gpc^{-3}\,yr^{-1}}$ for isotropic viewing angles. The isotropic limit lies above the entire GWTC-4.0 90% range for the BNS merger rate (7.6–$250\,\mathrm{Gpc^{-3}\,yr^{-1}}$). The polar-view limit falls within that range; it constrains only a population in which every merger produces a polar-viewed AT2017gfo-like kilonova. Over the same 23 months, our simulated sensitivity is higher than that inferred by Andreoni et al. (2020). Over eight years it grows 5.4× beyond that window. [TBD: Population-averaged limit; comparison including pipeline efficiency; final N_{obs}; implications for Rubin (e.g. Andreoni et al. 2022).]

Based on observations obtained with the Samuel Oschin 48-inch Telescope at Palomar Observatory as part of the Zwicky Transient Facility project. Computations were performed at the Minnesota Supercomputing Institute. [TBD: ZTF partnership and funding acknowledgments.]

Facilities: PO:1.2m

Software: `survey-sim`, `fiesta` (Koehn et al. 2025), `numpy`, `scipy`, `pandas`, `matplotlib`

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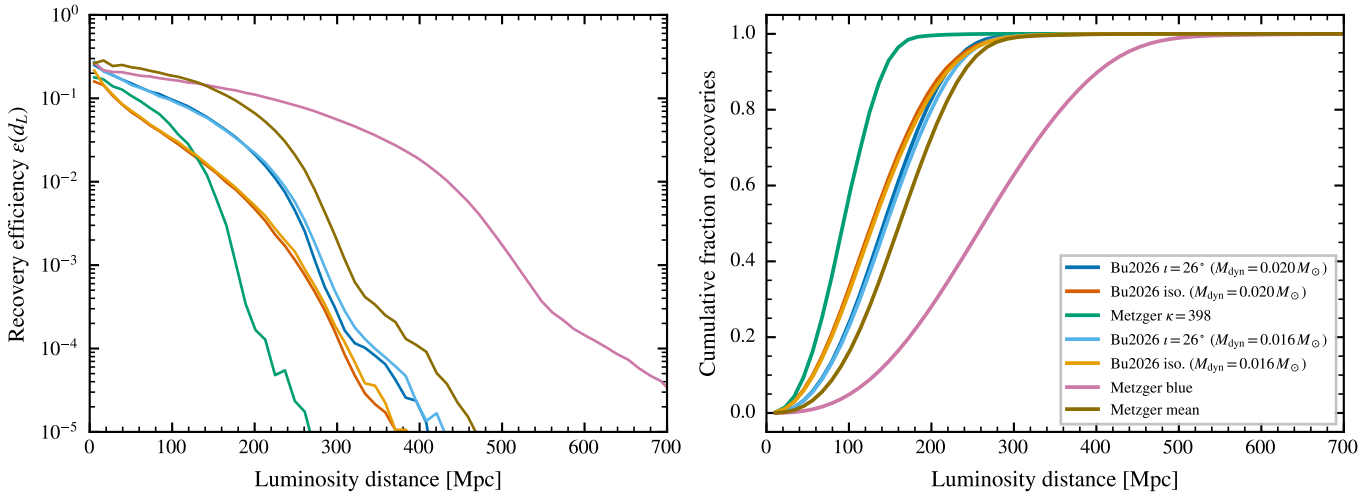


Figure 3. Left: recovery efficiency as a function of luminosity distance (recovered over injected injections per redshift bin; injections are uniform in comoving volume). Right: cumulative distribution of the luminosity distances of recovered injections for every model.

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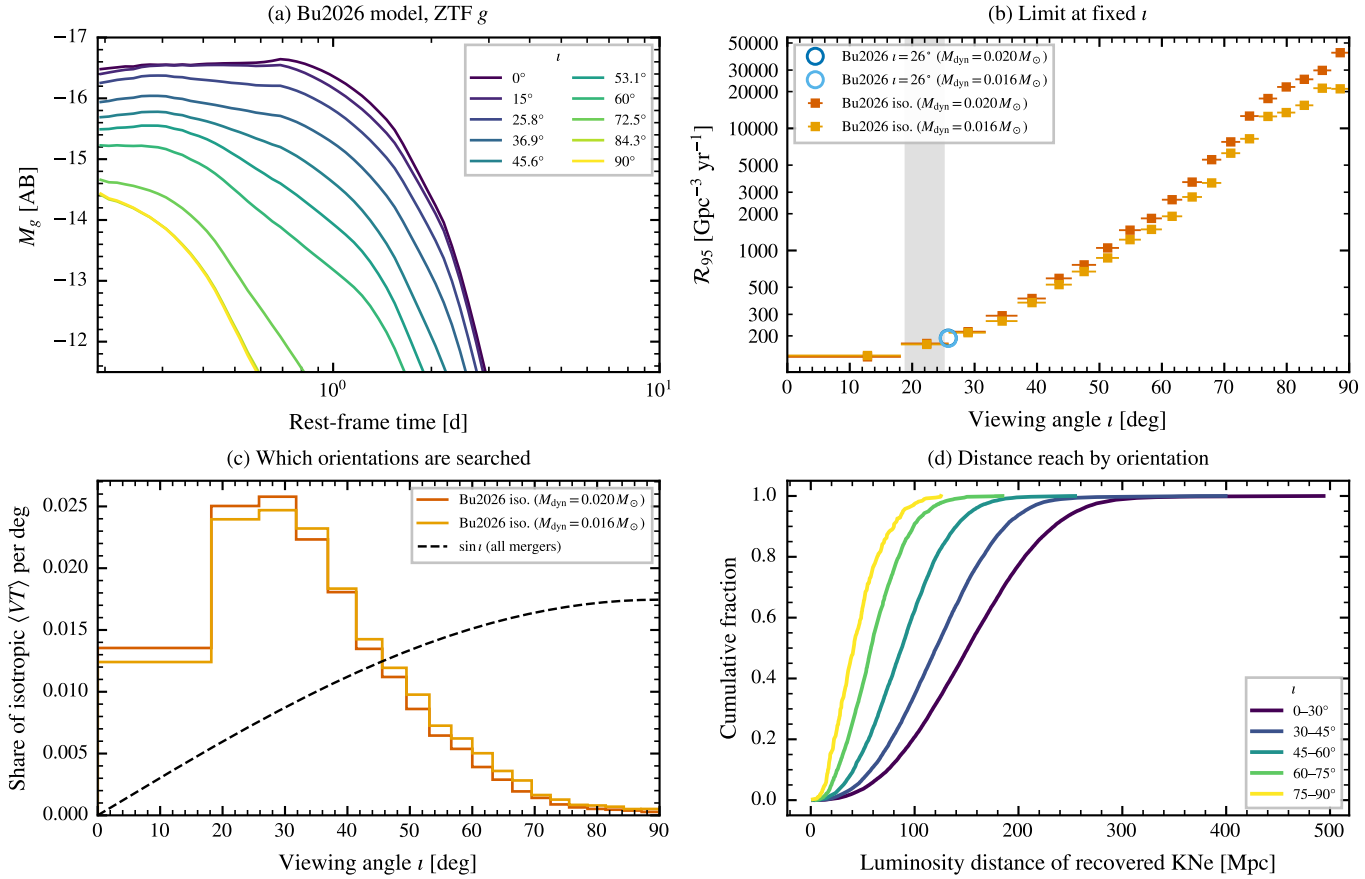


Figure 4. Viewing-angle dependence. (a) Rest-frame g-band light curves of the simulated ejecta from the POSSIS surrogate for viewing angles from pole-on to edge-on. (b) 95% limit for populations at fixed viewing angle, derived from the isotropic ensembles in 20 bins uniform in $\cos\iota$ (horizontal bars give the bin width); open circles are the independent fixed-angle (25.8°) ensembles; the grey band is the GW170817 viewing angle of Mooley et al. (2022). (c) Share of the isotropic $\langle VT\rangle$ contributed per degree of viewing angle, compared with the $\sin\iota$ distribution of random orientations (dashed). (d) Cumulative luminosity-distance distributions of recovered injections in five viewing-angle ranges (isotropic ensemble of the baseline set).

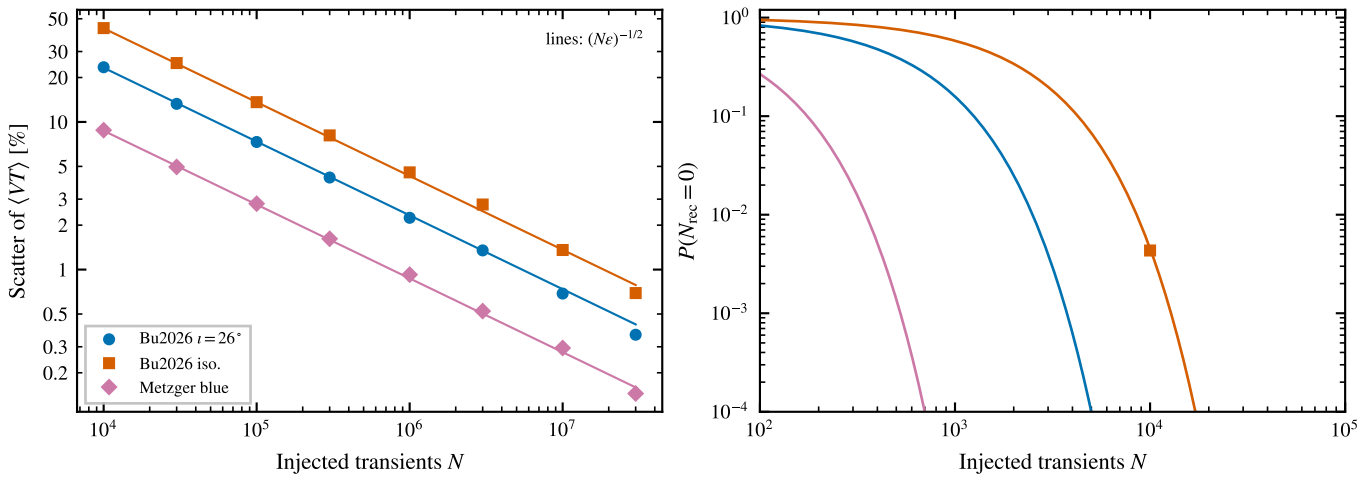


Figure 5. Left: relative scatter of $\langle VT\rangle$ as a function of the number of injections, from sub-sampling the simulated realisations (markers), compared with the binomial expectation (lines). Right: probability that a simulation recovers no injection, $e^{-\epsilon N}$ (lines), and the fraction of sub-samples with none (markers).